



HR Analytics Dashboard

(Employee Attrition Analysis using Power BI)



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DATA ANALYST PROJECT
POWER BI



PROJECT OVERVIEW



This project analyzes employee attrition patterns, workforce demographics, salary trends, and job satisfaction using Power BI.



Business Problem



- High employee attrition impacts company cost
- HR needs better workforce insights
- Identify departments with high turnover
- Understand employee behavior



Dataset Information



I got this dataset from **Kaggle.com**. This is a HR dataset. This **dataset has 1480 records**. I used Power BI for this project because it provide **powerful data visualizations, interactive dashboard** and **DAX capabilities** that **help transform new HR data into actionable business insights**.



Data Cleaning & Transformation



-
- Removed null values
 - Checked data types
 - Created DAX measures
 - Built calculated columns
 - Used Power Query



KPIs Used



1. Total Employees
4. Actual Employees

2. Total Attrition
5. Average Age

3. Attrition Rate
6. Average Salary

1480

Total Employees



238

Total Attrition



16%

Attrition Rate



1242

Active Employees



37

Average Age



6505

Average salary



HR ANALYTICS DASHBOARD

Human Resources

Research & Development

Sales

1480

Total Employees



238

Total Attrition



16%

Attrition Rate



1242

Active Employees



37

Average Age



6505

Average salary



Female

Male

Divorced

Married

Single

Non-Travel

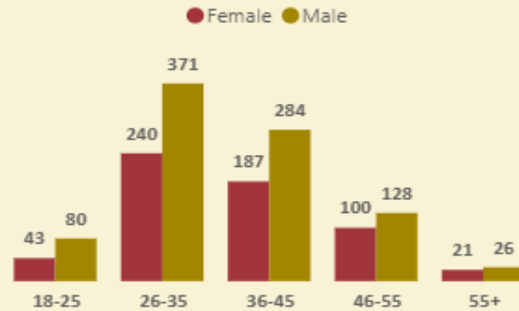
Travel_Frequently

Travel_Rarely

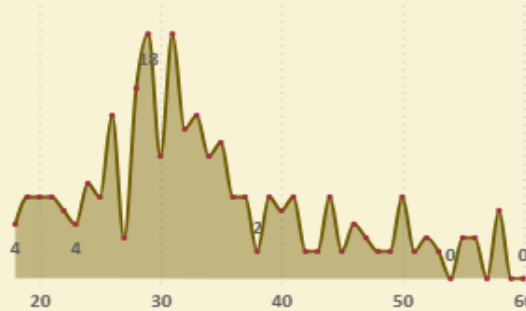
TravelRarely

Clear all slicers

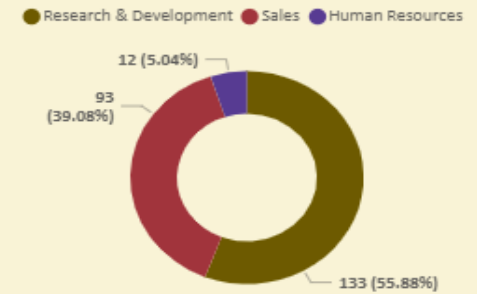
Employees by Age Group & Gender



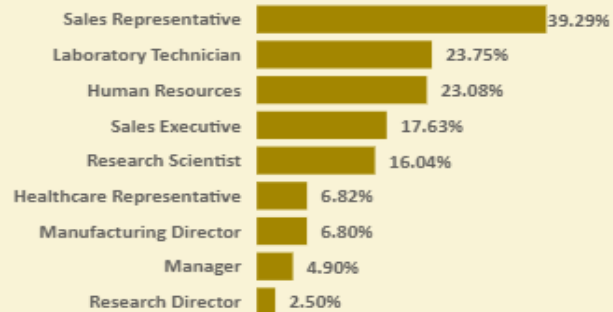
Attrition by Age



Department wise Attrition



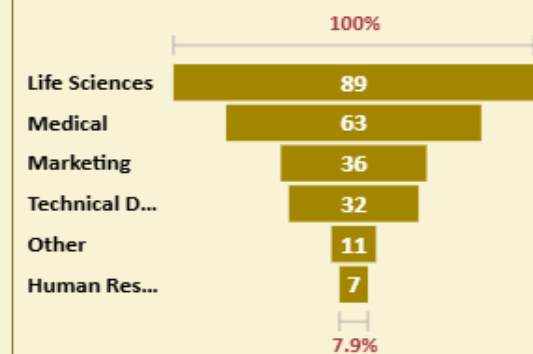
Attrition Rate by JobRole



Job Satisfaction Rating

JobRole	1	2	3	4	Total
Healthcare Representative	27	19	43	43	132
Human Resources	10	16	13	13	52
Laboratory Technician	56	48	76	81	261
Manager	21	21	27	33	102
Manufacturing Director	27	33	49	38	147
Research Director	15	16	27	22	80
Research Scientist	55	53	90	95	293
Sales Executive	70	55	91	113	329
Sales Representative	12	21	28	23	84
Total	293	282	444	461	1480

Attrition by Education Field





Key Insights



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1. High attrition in Sales and Research and Development Department.
 2. Age group between 26-35 have more attrition.
 3. Research & Development and Sales departments see the most employee turnover.
 4. Single employees show higher attrition compared to married and divorced counterparts.



Challenges Faced



1. The dataset contained unnecessary columns, inconsistent data types, and missing values that needed preprocessing before analysis.
2. Creating dynamic KPI measures such as Attrition Rate and Average Salary required proper understanding of DAX functions.
3. Maintaining proper alignment, spacing, and consistent formatting across visuals was challenging during dashboard design.
4. Choosing the right visualizations to represent HR insights effectively required careful analysis.
5. Managing multiple visuals and calculations while maintaining dashboard responsiveness was an important consideration.
6. The project improved my skills in data cleaning, DAX calculations, dashboard design, and business insight generation using Power BI.



Conclusion



The dashboard provides actionable HR insights that help improve employee retention strategies and support effective workforce management.



THANK YOU!!

